WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-07-20 to 2026-07-26

Curated articles in political communication and computational social science.

SUMMARY

This week's COMAP Journal Bot brief is especially strong at the intersection of computational media and institutional governance. The new set pairs close studies of political talk, propaganda, live-streamed emotional contagion, and AI-companion privacy with a substantial PNAS cluster on algorithms, law, disparate impact, and AI safety. It also surfaces useful methods work on linguistic polarization, conjoint measurement error, judicial citation networks, human mobility, and agentic data-science multiverses.

- Start with the multimodal debate study and the propaganda research agenda for the clearest political-communication throughline.
- The AI governance papers are unusually complementary: they move from collaboration design and benchmarking to legal interpretation, regulation, and disparate impact.
- Inequality appears across negotiation, mobility, environmental exposure, academic pay, policing, and intergenerational opportunity.

Collection completed at 2026-07-26 23:59 KST, just before the nominal Sunday window closed; publications indexed afterward can still be recovered as late additions in the next run.

HIGHLIGHTS

Affective Publics in the YouTube Live Chat: Multimodal Emotional Contagion During the 2024 U.S. Presidential Debate

POLITICAL COMMUNICATION · 2026-07-22

Why it matters · A rare end-to-end multimodal political communication study links candidates' facial and vocal emotion to real-time audience reactions in 53,095 live-chat messages.

Open article →

Studying Propaganda as a Social Process

POLITICAL COMMUNICATION · 2026-07-22

Why it matters · This intervention moves beyond isolated falsehoods and offers a broader research agenda for propaganda's rhetorical functions, persuasive appeal, and anti-democratic power.

Open article →

Chatting with confidants or corporations? Privacy management with AI companions

JOURNAL OF COMPUTER-MEDIATED COMMUNICATION · 2026-07-24

Why it matters · Interviews with AI-companion users show how anthropomorphic design lowers disclosure thresholds and entangles relational intimacy with corporate data governance.

Open article →

Polarization as a linguistic feature: a systematic review of computational operationalizations of polarization

COMMUNICATION METHODS AND MEASURES · 2026-07-26

Why it matters · The review directly addresses a recurring measurement problem in computational political communication: how polarization is operationalized in language.

Open article →

Capable language models can outgrow the benefits of collaboration

NATURE MACHINE INTELLIGENCE · 2026-07-24

Why it matters · A controlled comparison across 260 configurations offers a practical threshold for deciding when multi-agent coordination helps-and when it only adds overhead.

Open article →

Measuring disparate impact in human and machine decisions

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-07-20

Why it matters · The paper connects algorithmic decision-making to a central governance question: how to define and measure unequal effects across human and automated systems.

Open article →

Insights from climate psychology for misinformation research

NATURE HUMAN BEHAVIOUR · 2026-07-20

Why it matters · This cross-field perspective brings established insights from climate psychology into the design and interpretation of misinformation research.

Open article →

Many AI analysts, one dataset: Navigating the agentic data science multiverse

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-07-14

Why it matters · Autonomous analysts reproduce meaningful analytic dispersion, making a strong case for multiverse-style reporting and prompt disclosure in AI-assisted empirical research.

Open article →

TABLE OF CONTENTS

New This Week

- Communication Methods and Measures
- Journal of Computer-Mediated Communication
- Nature Machine Intelligence
- Proceedings of the National Academy of Sciences
- American Journal of Political Science
- Nature Communications
- Nature Computational Science
- Political Communication
- Nature Human Behaviour

Previous Week Catch-Up

- American Journal of Political Science
- Political Communication
- Science Advances
- Proceedings of the National Academy of Sciences
- American Political Science Review
- Journal of Communication
- Nature
- Nature Communications

Late Additions

- Nature

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
284	154	128
LATE ADDITIONS CANDIDATES	**CURATED DIGEST BELOW**	**NOTE**
2	24 new, 13 catch-up, 1 late addition	The full collected set above is before manual relevance filtering, while the attached PDF reflects the curated digest.

FULL CURATED DIGEST

New This Week

Communication Methods and Measures

Polarization as a linguistic feature: a systematic review of computational operationalizations of polarization

2026-07-26 · DOI 10.1080/19312458.2026.2704557

Helena Rauxloh

GESIS Leibniz Institute for the Social Sciences

Abstract. Abstract unavailable.

Read article →

Journal of Computer-Mediated Communication

Chatting with confidants or corporations? Privacy management with AI companions

2026-07-24 · DOI 10.1093/JCMC/ZMAG014

Hsuen-Chi Chiu; Jeremy Foote

Brian Lamb School of Communication, Purdue University , West Lafayette, IN,

Abstract. Abstract AI companion chatbots feel like intimate confidants but exist as part of corporate data systems. Drawing on Communication Privacy Management (CPM) theory and Masur's dimensional privacy framework, we interviewed users of platforms like Replika and Character.AI (N = 15) to understand their privacy decisions in AI companion interactions. We find that anthropomorphic design lowers disclosure thresholds, encouraging increasingly intimate self-disclosure. As relationships develop, some users engage in simulated co-ownership, treating shared information as relationally held despite recognizing corporate control. Because post-disclosure correction feels futile, privacy management becomes anticipatory and front-loaded-users rely on distancing strategies such as using fake names rather than post-hoc deletion. For some users, privacy turbulence centers less on fears of exposure and more on fears of losing conversational memory-a relational reset feels more threatening than a data breach. These findings extend CPM theory by showing that in human-AI companionship contexts, relational and institutional privacy concerns become inextricably entangled.

Read article →

Nature Machine Intelligence

Capable language models can outgrow the benefits of collaboration

2026-07-24 · DOI 10.1038/S42256-026-01268-Y

Yubin Kim; Ken Gu; Chanwoo Park; Chunjong Park; Samuel Schmidgall; A. Ali Heydari; Yao Yan; Zhihan Zhang; Yuchen Zhuang; Yun Liu; Mark Malhotra; Paul Pu Liang; Hae Won Park; Yuzhe Yang; Xuhai Xu; Yilun Du; Shwetak Patel; Tim Althoff; Daniel McDuff; Xin Liu

Abstract. Agents, language model-based systems that can reason, plan and act with tools to accomplish tasks, are widely deployed, yet it remains unclear when multi-agent coordination outperforms a strong single agent. Here we conduct a controlled experiment that holds task prompts, tools and compute budgets constant while varying only coordination structure

and model capability. Across 260 configurations spanning six benchmarks, five architectures and three LLM families, we derive a predictive model using empirical coordination metrics. Across benchmarks, single-agent baseline performance emerges as the most robust predictor of whether coordination improves or decreases performance. In particular, we identify an empirical capability-saturation threshold beyond which additional agents are unlikely to improve performance. This threshold correctly predicts the effect of multi-agent coordination on performance in 94% of validation configurations on SWE-bench Verified and Terminal-Bench. We therefore interpret this threshold as a practical selection rule rather than a universal scaling principle. A second effect, baseline-scaled error amplification, survives cluster-robust inference (Probust = 0.030) and supports the failure-mode taxonomy. The fitted model achieves cross-validated R2 = 0.373 (0.413 with a task-grounded capability metric) and selects the best architecture in 87% of held-out configurations. These results provide a quantitative framework for within-domain architecture selection and for estimating when multi-agent coordination is likely to improve performance or add overhead. A controlled study of large language model agents across 260 configurations shows when multi-agent collaboration helps or hurts performance, and introduces a predictive model that selects the best architecture in 87% of held-out within-domain configurations.

Read article →

Proceedings of the National Academy of Sciences

The social-class gap in negotiation: Lower-class individuals negotiate less and face more backlash

2026-07-24 · DOI 10.1073/PNAS.2602128123

Ying Lin; Jackson G. Lu; Michele J. Gelfand

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Abstract. Economic inequality has widened considerably in the United States and globally, raising pressing questions about the mechanisms that reproduce social stratification. One prominent manifestation of this inequality is the "class ceiling" in organizations: individuals from lower social classes earn less and advance more slowly than their higher-class counterparts, even with comparable credentials. We identify a crucial behavioral pathway contributing to this class ceiling: a social-class gap in the propensity to negotiate. We propose that lower-class individuals are less likely to initiate negotiations than their higher-class counterparts, a pattern that can compound economic disadvantage over time. Across four studies (total N = 10,211)-including a nationally representative sample of employees (Study 1), a sample of business school students (Study 2), a field study of an online labor market (Study 3), and a survey study of workplace negotiations and career outcomes (Study 4)-we find consistent evidence for this gap. This gap is partially explained by lower-class individuals' reduced sense of power and heightened concerns about social backlash. Additionally, Study 5 (N = 1,133) finds that such concerns may be justified: an experiment with human resources (HR) professionals shows that identical negotiation requests elicit stronger social backlash when initiated by lower-class (vs. higher-class) individuals. Together, these findings reveal a pernicious double bind for lower-class individuals: remaining silent perpetuates economic disadvantage, yet initiating negotiation exposes them to greater social penalties. Addressing this double bind is critical for ensuring that negotiation serves as a vehicle for upward mobility rather than a gatekeeper of privilege.

Read article →

Continuity and change in US legal tradition: Evidence from judicial citation communities

2026-07-20 · DOI 10.1073/PNAS.2509763123

Elliott Ash; Caterina Chiopris; Robert Mahari; Suresh Naidu

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Abstract. Legal doctrine is often described as evolving case by case while preserving coherent legal traditions. We introduce a method for identifying and tracing such traditions by applying the Louvain community-detection algorithm to the complete

U.S. federal judicial citation graph (1790-2024; 760,000 opinions linked by 8.3 million citations). The resulting citation communities are unsupervised, blind to court labels and topical taxonomies, yet they form meaningful empirical groupings: In validation exercises, they recover known institutional structure and capture coherent substantive and procedural approaches to law. Using these communities as empirical units, we study how legal traditions evolve and respond to legislative change. We show that the introduction of pivotal federal statutes shifts citation patterns toward newer precedent, indicating that legislation can disrupt established traditions and redirect the development of case law.

Read article →

Dietary DNA in municipal wastewater reveals signatures of wealth, immigration, and coastal proximity

2026-07-20 · DOI 10.1073/PNAS.2530704123

Mengyi Dong; Thomas Joseph Clerkin; Sharon Jiang; Nolan Ives; Olivia W. Osborne; Michelle Kirtley; Anna E. Bauer; Katherine Y. Anderson; Martin D. Smith; Rachel T. Noble; Lawrence A. David

Department of Environmental Sciences and Engineering, Gillings School of Global Public Health, University of North Carolina at Chapel Hill; Department of Molecular Genetics and Microbiology, Duke University; Duke Microbiome Center, Department of Molecular Genetics and Microbiology, Duke University; Institute of Marine Sciences, as part of the Department of Earth, Marine and Environmental Sciences, University of North Carolina at Chapel Hill; Nicholas School of the Environment and Department of Economics, Duke University

Abstract. Public health nutrition lacks scalable, objective tools for real-time dietary surveillance. We developed FoodSeq-FLOW (Food Landscape Observation in Wastewater), a genomic platform that sequences chloroplast trnL and mitochondrial 12SV5 DNA in municipal wastewater. Across 183 samples from 21 North Carolina wastewater treatment plants serving 2.1 million people, we detected 184 plant and 116 animal food taxa at a cost of

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Gender pay gaps in the social sciences

2026-07-20 · DOI 10.1073/PNAS.2524119123

Ayelet Gneezy; Uri Gneezy; Joshua Graff Zivin; Gaurav Khanna; Elizabeth Lyons; Marta Serra-Garcia

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Abstract. We provide a longitudinal, multifield analysis of the gender pay gap in academia that directly links individual-level salaries to measures of scholarly productivity. Using administrative data from the ten campuses that comprise the University of California system, we combine detailed pay records with bibliometric data on publications and citations for faculty in five major social science fields. We find a raw gender pay gap of 23%, which decreases to 4.3% after controlling for field, campus, and cohort. Including controls for job title and scholarly productivity has little effect on the estimated gap, although they substantially increase explanatory power. A decomposition shows that 17% of the gap is unexplained, persisting across junior and senior faculty. We also find a large, persistent gender gap in publications, but no citation gap once output and career length are considered. Field-level analysis reveals that pay gaps are not universal: They persist in Business, Sociology, and Anthropology, but are absent in Economics, Political Science, and smaller social science fields. Our results suggest that pay transparency is insufficient to eliminate salary disparities and that more equal representation within the field is not a prerequisite for pay equity. Our study isolates the residual gap that remains unexplained by productivity or seniority, challenging the view that these factors fully account for gender pay differences. Taken together, our findings highlight the role of field-specific norms and institutional structures, offering directions to promote equitable compensation in academia.

Read article →

Legal infrastructure for transformative AI governance

2026-07-20 · DOI 10.1073/PNAS.2509742123

Gillian K. Hadfield

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Abstract. Most of our AI governance efforts focus on substance: What rules do we want in place? What limits or checks do we want to impose on AI development and deployment? But a key role for law is not only to establish substantive rules but also to establish legal and regulatory infrastructure to generate and implement rules. The transformative nature of AI calls especially for attention to building legal and regulatory frameworks. In this Perspective, I review three examples: the creation of registration regimes for frontier models; the creation of registration and identification regimes for autonomous agents; and the design of regulatory markets to facilitate a role for private companies to innovate and deliver AI regulatory services.

Read article →

Measuring disparate impact in human and machine decisions

2026-07-20 · DOI 10.1073/PNAS.2509765122

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Abstract. Empirical analyses have grown increasingly important in discrimination litigation with the greater availability of detailed data on individuals and decisions. A popular analytic strategy is to estimate disparities after adjusting for observed covariates, typically with a regression model, in hopes of ferreting out discriminatory intent. This approach, however, is ill-suited to auditing algorithms that are now commonly used to aid decisions, which typically do not include race or other legally protected factors as inputs. Motivated by legal understandings of disparate impact, we introduce an approach that aims to measure "unjustified" disparities in both human and machine decisions. Our method, which we call risk-adjusted regression, proceeds in three steps. In the first step, we combine all available information in a machine learning model to estimate the value, or inversely, the risk, of taking a certain action, such as approving a loan application or hiring a job candidate. Second, we measure disparities in decisions after adjusting for these risk estimates alone. Finally, in the third step, we assess the sensitivity of results to potential mismeasurement of risk. We demonstrate this approach on a detailed dataset of 2.2 million police stops of pedestrians in New York City, and show that traditional statistical tests of discrimination can substantially understate the magnitude of (risk-adjusted) racial disparities.

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Public responses to atrocities depend on partisanship and context, not legal labels

2026-07-20 · DOI 10.1073/PNAS.2608203123

Yphtach Lelkes; Ellie M. Mitchell; Erin Walk; Sean J. Westwood

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Abstract. Legal labels such as genocide and crimes against humanity carry substantial legal and diplomatic weight in elite and institutional arenas, where classification shapes treaty obligations, ICC jurisdiction, and the perceived stakes of inaction. A common assumption in advocacy and policy discourse is that these labels also shape mass public judgment. Direct evidence is limited and ambiguous. We test this assumption in a preregistered, nationally representative survey experiment (N = 2 , 000) comparing four legally salient atrocity terms-genocide, war crimes, crimes against humanity, and ethnic cleansing-against a nonlegal control across five real-world conflict contexts, measuring support for punitive and humanitarian policies as well as moral attributions of responsibility. Across outcomes, estimated label effects are small and rarely distinguishable from zero. Critically, even within a given partisan and country context, swapping legal labels produces smaller shifts than the baseline gaps across partisan and geopolitical contexts-indicating that where you stand and who you are matter far more than what word is used. Any departures from zero are narrow and do not accumulate into a consistent pattern across outcomes or conflicts. These results suggest that the political significance of atrocity labeling lies primarily in elite and

institutional arenas; for mass publics, legal terminology is a weak lever compared to country context and partisan commitments.

Read article →

Race-conscious admissions algorithms and the law

2026-07-20 · DOI 10.1073/PNAS.2509764123

Alexandra Chouldechova; Daniel J. Hemel

Microsoft Research New York, Microsoft Corporation; New York University School of Law

Abstract. In recent years, colleges and universities have begun to use machine learning (ML) systems to inform admissions decisions. Meanwhile, in the 2023 case Students for Fair Admissions, Inc. v. President and Fellows of Harvard College , the Supreme Court held that colleges and universities may not make admissions decisions "on the basis of race." These parallel developments-the rise of ML in admissions and the fall of race-based affirmative action-will force educational institutions, and ultimately courts, to confront the difficult question of what it means for ML systems to differentiate "on the basis of race." We begin by mapping the Students for Fair Admissions decision onto different uses of race in predictive AI. We distinguish between "first-order" and "second-order" race consciousness at both the training and predictive phases of machine learning, and we argue that each category of race consciousness raises distinct legal and normative issues. We go on to show that the Students for Fair Admissions decision potentially permits-and even endorses-certain forms of race consciousness. Our analysis is grounded in the observation that the process of developing ML-based systems enables policymakers to calibrate decision making algorithms much more precisely and explicitly in response to specific criticisms of race-conscious affirmative action.

Read article →

Statutory construction and interpretation for AI

2026-07-20 · DOI 10.1073/PNAS.2509766123

Luxi He; Nimra Nadeem; Michel Liao; Howard Chen; Danqi Chen; Mariano-Florentino Cuellar; Peter Henderson

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Abstract. AI systems are increasingly governed by natural language rules, yet a key challenge arising from reliance on language remains underexplored: interpretive ambiguity. As in legal systems, ambiguity arises both from how these rules are written and how they are applied. But while legal systems use institutional safeguards to manage such ambiguity, such as transparent appellate review policing interpretive constraints, rule-based AI alignment pipelines lack comparable protections. Different interpretations of the same rule can lead to inconsistent model behavior. Drawing on U.S. legal theory, we identify key gaps in current rule-based alignment pipelines by examining how legal systems constrain ambiguity at both the rule creation and rule application steps. We then propose a computational framework that formalizes interpretive ambiguity as constrained entropy minimization and introduces two law-inspired mechanisms: 1) a rule refinement pipeline that minimizes interpretive disagreement by revising ambiguous rules (analogous to agency rulemaking or iterative legislative action), and 2) prompt-based interpretive constraints that reduce disagreement during rule application (analogous to legal canons that guide judicial discretion). We evaluate our framework on a 5,000-scenario subset of the WildChat dataset and show that both interventions significantly improve judgment consistency across a panel of reasonable interpreters. Our approach offers a step toward systematically managing interpretive ambiguity, an essential step for building more robust, law-following AI systems.

Read article →

The backfiring effect of weak AI safety regulation

2026-07-20 · DOI 10.1073/PNAS.2509768123

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Abstract. Recent policy proposals aim to improve the safety of general-purpose AI, but there is little understanding of the efficacy of different regulatory approaches. We present a strategic model that explores interactions between safety regulation, general-purpose AI technology creators, and domain specialists-those who adapt the technology for specific applications. Our analysis examines how regulatory measures targeting different parts of the AI development chain affect the outcome of this game. Our model assumes AI technology is characterized by two key attributes: safety and performance. The regulator first sets a minimum safety requirement that applies to one or both players. The general-purpose creator then invests in the technology, establishing its initial safety and performance levels. Next, domain specialists refine the AI for their use cases, updating the safety and performance levels and taking the product to market. Resulting revenue is shared between the specialist and generalist. Our analysis reveals two insights: first, weak safety regulation imposed predominantly on domain specialists can backfire. While it might seem logical to regulate AI use cases, our analysis shows that weak regulations targeting domain specialists alone can reduce safety in a large class of parameterizations. Second, in contrast to the previous finding, we observe that stronger, well-placed regulation can mutually benefit all players. When regulators impose appropriate safety standards on both general-purpose AI creators and domain specialists, the regulation can function as a commitment device, leading to safety and performance gains, surpassing what is achievable under no regulation or regulating only one player.

Read article →

There is no free benchmark: An institutional view of legal AI benchmarking

2026-07-20 · DOI 10.1073/PNAS.2509757122

Neel Guha; Andy K. Zhang; Christine Tsang; Christopher D. Manning; Julian Nyarko; Daniel E. Ho

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Abstract. Despite substantial excitement around the use of AI in law, little information exists on the performance and associated risks of the domain's widely marketed tools. Recent work, for instance, has demonstrated the significant potential for "hallucinations"-wherein models make up facts, law, and precedent-leading Chief Justice Roberts to spotlight this risk in his annual report on the judiciary. We argue that there is a need for public AI benchmarking in law. First, relative to other AI application domains, the legal AI ecosystem lacks legibility-there is little information about the design and performance of many commercial legal AI systems. Legal AI has not benefited from the types of benchmarking that have catalyzed, measured, and informed AI innovation and responsible use in other domains. Second, we articulate the challenges of the institutional design of benchmarking. We illustrate how benchmarks can be captured, watered down, and abused. Careful institutional design around the why, who, what, and how of benchmarking will be critical to navigate difficult tradeoffs of transparency, objectivity, expertise, and resources. Third, addressing legal AI's illegibility requires matching institutional models to available resources and constraints. Rather than advocating for a single "best" approach to benchmarking, we show how benchmarking strategies depend on available resources.

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American Journal of Political Science

Correcting measurement error bias in conjoint survey experiments

2026-07-23 · DOI 10.1111/AJPS.70076

Katherine Clayton; Yusaku Horiuchi; Aaron R. Kaufman; Gary King; Mayya Komisarchik

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Abstract. Abstract Conjoint survey designs are spreading across the social sciences due to their unusual capacity to estimate many causal effects from a single randomized experiment. Unfortunately, by their ability to mirror complicated realworld choices, these designs often generate substantial measurement error and thus bias. We replicate both the data collection and analysis from eight prominent conjoint studies, all of which closely reproduce published results, and reveal high levels of measurement error in all. We then discover a common empirical pattern in how measurement error appears in conjoint studies and, with it, introduce an easytouse statistical method to correct the bias. Along the way, we provide a much simpler and simultaneously more powerful approach to designing, organizing, understanding, and analyzing conjoint data analyses.

Read article →

Nature Communications

Disproportionate exposure to extreme floods in historically redlined U.S. communities

2026-07-23 · DOI 10.1038/S41467-026-76024-2

Elizabeth Appel; S. Amin Enderami; Elaina J. Sutley; Admin Husic

Abstract. Abstract Extreme floods and historical public policies both leave lasting impacts. We analyzed flood exposure across the neighborhoods of 202 US cities historically assessed for lending risk by the Home Owners' Loan Corporation on a scale from 'best (A) to 'hazardous' (D). Using high-resolution flood modeling and dasymetric population mapping, we found that D-graded neighborhoods bear a disproportionate flood burden in roughly three-quarters of cities. This inequality follows a systematic gradient: A- and B-graded neighborhoods are underrepresented in the flooded population, C-graded neighborhoods are proportionately represented, and D-graded neighborhoods are overrepresented. A greater fraction of D-graded residents are exposed to the 100-yr flood (6.4%) than A-graded residents to the 1000-yr flood (5.9%) with the gap widening at higher-magnitude floods. Flood disparities are most pronounced in the Midwest and Mid-Atlantic regions, aligning with the historical concentration of industrial labor and redlined neighborhoods along waterways. However, we find that proximity to waterways alone does not explain the observed inequality: flood disparities persist even among neighborhoods far from major rivers and coasts, implicating infrastructure deficits as a compounding factor. Neighborhoods marked as 'hazardous' nearly a century ago remain disproportionately vulnerable today, underscoring the need for equitable urban planning.

Read article →

Structural inequality shapes the daily rhythm of stationarity and mobility

2026-07-23 · DOI 10.1038/S41467-026-75778-Z

Zihao Tian; Chao Zhang; Weiyu Li; Lixin Tian; Michael M. Danziger; Wenshan Liu; Qi R. Wang

Abstract. Human mobility exhibits rhythmic patterns that shape economic activity, social interactions, and access to essential services. Yet, while prior studies focus on spatial movement, the temporal structure of people's stationarity-the act of dwelling-remains underexplored. Here, we introduce the Stationarity Sinusoidal Stability (STASIS) model to quantify the daily pulse of physical presence. Using three parameters-range, phase angle, and baseline-we show that majority-Black census block groups (CBGs) have smaller daily stationarity ranges, later activity peaks, and lower average home-presence baselines than majority-White and majority-Hispanic CBGs. The Rhythm Stability Index further indicates greater variability in activity timing among CBGs with higher proportions of Black residents, and this variability is associated with CBG-level economic precarity. Using COVID-19 as a stress test, our findings show that temporal stability in daily mobility rhythms differs across

CBGs with different racial and Hispanic-origin compositions. The authors model daily routines of rest and activity, revealing that majority-Black communities experience highly volatile schedules. This shows that temporal predictability is a fundamental yet overlooked dimension of social justice.

Read article →

Adult-to-infant unidirectional neural coupling mediates selective social learning in infants from the United Kingdom and Singapore

2026-07-22 · DOI 10.1038/S41467-026-75831-X

Wei Zhang; Kaili Clackson; Stanimira Georgieva; Lorena Santamaria; Vanessa Reindl; Valdas Noreika; Nicholas Darby; Vaka Valsdottir; Priyadharshini Santhanakrishnan; Victoria Leong

Abstract. Infants possess powerful learning abilities, but their neural resources must be deployed parsimoniously toward the most relevant information in a given social context. Adults curate infants' information selection through ostensive marking of valuable content using social cues such as eye-contact. However, the neural mechanisms that support social gating of early learning remain unclear. Here, we demonstrate that - in infants from UK and Singapore - unidirectional adult-speaker to infant-listener neural coupling mediates infants' selection of socially-relevant information for learning. Specifically, speaker-to-listener neural coupling is modulated by the adult speaker's eye contact and predicts selective learning of an artificial language that is paired with full (compared to partial or no) eye contact. Speaker-to-listener neural coupling is a better predictor of selective learning than infants' own neural activity, suggesting that this signal effectively captures important social influences on early cognition, and may provide insights into social valuation decision-processes that dictate early social learning. Infants are sensitive to social cues like eye gaze as a signal for learning. Here, the authors show, in cohorts from Singapore and the United Kingdom, coupling of adult and infant brain activity mediates infants' use of speaker gaze to learn selectively.

Read article →

Emerging capitalism and sociocultural niche construction in early modern Greece

2026-07-22 · DOI 10.1038/S41467-026-75315-Y

Georgios C. Liakopoulos; Marcus Gro; Piotr Guzowski; Katerina Kouli; Dimitrios Lamprakis; Alessia Masi; Theodoros Vakkas; Elena Xoplaki; Adam Izdebski; Ricardo Fernandes

Abstract. Abstract Sociocultural niche construction is a key mechanism behind the current planetary crisis. Its causal drivers in past societies remain poorly understood due to a lack of data and modeling complexities. Here, we apply a Bayesian machine learning algorithm to a geospatial quantitative dataset compiled from 15 th 16 th c. CE Ottoman taxation registers, to explore shifts in drivers of land use choices in village communities in southern Greece. Greek and Albanian communities living within the same landscape experienced a transition from unstable political conditions and limited commodification of agriculture to lasting peace, intense market integration and state fiscal pressure, characteristic for early European capitalism. We found that prior to this economic transformation, ethno-cultural identities (Greek or Albanian) played an important role in local ecological niche construction. However, within two human generations (50 years), village communities shifted away from relying on ethnic identities in their ecological choices and their strategies became strongly connected to the investment potential of each village community and the environmental characteristics of their surroundings. Thus, our analysis demonstrates that the pressures of early modern socio-economic systems necessitated highly adaptive behavior from local communities that reduced the ecological significance of traditional identities.

Read article →

Nature Computational Science

Rethinking privacy-preserving GWAS benchmarks for governance

2026-07-23 · DOI 10.1038/S43588-026-01023-8

Natnatee Dokmai

Abstract. Abstract unavailable.

Read article →

Political Communication

Common Identities and Political Talk: Evaluating Cross-Cutting Political Discussions in Sports Communities on Reddit

2026-07-23 · DOI 10.1080/10584609.2026.2704899

Ashwin Rajadesingan

University of Texas

Abstract. Research suggests that online nonpolitical communities may facilitate better cross-cutting political discussions than explicitly political communities as users' shared nonpolitical identities can offset partisan bias, in line with the Common Ingroup Identity Model. This study tests this idea through a large-scale computational analysis comparing the quantity, tone, and substance of political discussions on the same topics in political and sports communities on Reddit. Overall, we find that sports communities facilitate more cross-cutting political discussions than political communities, with the exception of soccer communities. These discussions use similar proportions of credible sources but use significantly fewer sources that are not credible compared to discussions in political communities. On the flip side, we do not find evidence of a meaningful difference in toxicity levels of cross-cutting political discussions in sports and political communities. Further, we find that political discussions in sports communities use fewer sources overall. These results advance a nuanced understanding of the promise of nonpolitical spaces for deliberative democracy and the role of common identities in political discussions.

Read article →

Affective Publics in the YouTube Live Chat: Multimodal Emotional Contagion During the 2024 U.S. Presidential Debate

2026-07-22 · DOI 10.1080/10584609.2026.2704891

Sungwon Jung; Zhi Lin; Joseph M Stepniewski; Jennifer Castillo Cortes; Yuzhe Lei

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Abstract. Televised debates between political candidates have become dynamic, interactive events on live-streaming platforms where audiences form affective publics through real-time interactions and emotional expressions in live chat. In this study, we focused on real-time emotional contagion on a right-wing YouTube channel during the live-streamed 2024 U.S. presidential debate between Donald Trump and Kamala Harris. Examining how multimodal emotional expressions from candidates influence audience responses in live chats (N = 53,095), this study introduces a novel, multi-dimensional approach combining computational techniques, including face emotion recognition, speech emotion analysis, text emotion classification, and time series analysis, to analyze multimodal emotional cues across audio, video, and chat data. Results revealed that Harris' vocal fear Granger-caused an increase in joy in the live chat, and that emotional contagion patterns varied by user engagement level and political leaning: low-engagement users were primarily influenced by the candidates, whereas higher-engagement users were more influenced by peer emotions. Pro-Trump audiences' joy was amplified by Harris's vocal sadness; among pro-Harris audiences, anger increased in response to Harris's vocal sadness and the moderators' facial sadness, while joy increased in response to Trump's facial sadness. Negative public emotions including sadness and anger also reinforced joy in the live chat. These findings offer new insights into the role of emotional contagion in mobilizing online affective publics during live-streamed political events.

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Studying Propaganda as a Social Process

2026-07-22 · DOI 10.1080/10584609.2026.2706643

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Abstract. In recent scholarship on propaganda, no aspect has attracted more attention than the digitally mediated propagation of disinformation. In this intervention, we argue that focusing narrowly on disinformation risks missing the forest for all the trees: Only by considering disinformation's embedding within broader propaganda strategies and processes, we can understand falsehoods' distinctive rhetorical functions alongside other forms of (normative, motivational, etc.) propaganda messaging, comprehend their persuasive appeal, and theorize their capacity to gain discursive power within contemporary high-choice information environments. Advancing an approach that foregrounds propaganda's inherent anti-democratic ambition over its manipulative means, we sketch a research agenda suitable to studying propaganda as a social process.

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Nature Human Behaviour

Insights from climate psychology for misinformation research

2026-07-20 · DOI 10.1038/S41562-026-02530-4

Tobia Spampatti; Bojana Veckalov; Sandra J. Geiger; Sander van der Linden; Stephan Lewandowsky; Matthew J. Hornsey

Abstract. Abstract unavailable.

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Previous Week Catch-Up

American Journal of Political Science

Contesting repression in divided democracies: Can counternarratives reduce support for police violence?

2026-07-19 · DOI 10.1111/AJPS.70073

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Abstract. Abstract Democracies often struggle to contain police violence, which undermines the rule of law and reproduces inequality by targeting marginalized citizens. Yet, scholars have shown that repressive policing enjoys considerable public support. Focusing on Brazil, we demonstrate that attitudes toward police violence mirror polarization between left and rightwing voters, and investigate how competing narratives change attitudes toward police violence through multiple mechanisms. We argue that marginalized victims' counternarratives can counteract police discourses justifying violence through perspectivegetting, enabling citizens to hear about the experiences of stigmatized groups. Using a nationwide survey experiment, we find that although police narratives increase endorsement of lethal violence, statements from victims' mothers emphasizing innocence and disadvantage reduce support for police killings-even among rightwing voters-by increasing empathy and reducing victimblaming. We demonstrate that, even in deeply unequal and polarized democracies where exclusionary attitudes align with entrenched ideological divides, counternarratives from marginalized groups can shape public

attitudes toward injustices.

Read article →

Electoral contests over voter beliefs

2026-07-19 · DOI 10.1111/AJPS.70075

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Abstract. Abstract Voter perceptions of candidates are the substrate on which campaigns act. It is costly, however, for voters to devote enough attention to politics to change their own perceptions of candidates. This paper views campaigns-and the time, effort, and money they expend-as subsidizing rationally inattentive voters' adoption of new beliefs about candidate platforms. We employ an allpay contest to model costly competition over voter perceptions. In our baseline model, campaigns seek to improve perceptions of their own candidates. In a variant, candidates engage in negative campaigning. Among other results, we observe platform divergence under more reasonable informational assumptions than previous spatial models, replacing candidate uncertainty about voters with voter uncertainty about candidates. This model serves as a bridge between behavioral research on campaigns and theoretical literature on elections, offering a richer set of empirical predictions while preserving the insights of the canonical Downsian framework as a limiting case.

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Political Communication

Political Fact-Checking Efforts are Constrained by Deficiencies in Coverage, Speed, and Reach

2026-07-19 · DOI 10.1080/10584609.2026.2704896

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Abstract. Fact-checking has been promoted as a key method for combating political misinformation. Comparing the spread of election-related misinformation narratives along with their relevant political fact-checks, this study provides the most comprehensive assessment to date of the real-world limitations faced by political fact-checking efforts. To examine barriers to impact, this study extends recent work from laboratory and experimental settings to the wider online information ecosystem present during the 2022 U.S. midterm elections. From analyses conducted within this context, we find that fact-checks as currently developed and distributed are severely inhibited in election contexts by constraints on their i. coverage, ii. speed, and, iii. reach. Specifically, we provide evidence that fewer than half of all prominent election-related misinformation narratives were fact-checked. Within the subset of fact-checked claims, we find that the median fact-check was released a full four days after the initial appearance of a narrative. Using network analysis to estimate user partisanship and dynamics of information spread, we additionally find evidence that fact-checks make up less than 1.2\% of narrative conversations and that even when shared, fact-checks are nearly always shared within,rather than between, partisan communities. Furthermore, we provide empirical evidence which runs contrary to the assumption that misinformation moderation is politically biased against the political right. In full, through this assessment of the real-world influence of political fact-checking efforts, our findings underscore how limitations in coverage, speed, and reach necessitate further examination of the potential use of fact-checks as the primary method for combating the spread of political misinformation.

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Reintegrating Influencer Media Logic into the Study of Political Influencers

2026-07-18 · DOI 10.1080/10584609.2026.2706641

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Abstract. Political influencers are an emerging epistemic authority in politics. Yet, existing scholarship risks conceptually misspecifying them by focusing on their political nature and neglecting their communicative practices established in broader social media influencer literature. This article advances a theoretical intervention to reintegrate influencer studies into the study of political influencers in political communication. Drawing on the media logic tradition, we discuss "influencer media logic" as communication principles that distinguish political influencers from other political media actors and that underpin their persuasiveness. We discuss three principles of influencer media logic, authenticity, intimacy, and platform expertise, and their contribution to restructuring visibility, credibility, and trust in politics. By foregrounding these communicative practices, we address the limitations of existing conceptualizations and clarify mechanisms of influence beyond the political aspect of their content. We discuss research directions to systematically study political influencers and examine their implications for political communication and democratic engagement.

Read article →

Vengeance or Hope? A Role-Based Analysis of Hashtag Activism for the War in Gaza

2026-07-14 · DOI 10.1080/10584609.2026.2702539

Tay Jeong

Sungkyunkwan University

Abstract. Using cross-lingual international X trends data and a unique role-based analytical framework well-suited for the study of granular textual data, this study examines the partisan and spatial distribution of political rhetoric in hashtag activism about the Gaza War in its early phase. Following a triadic classificatory rubric of roles composed of victim-villain-hero, I find that pro-Palestine hashtag activism was overwhelmingly focused on victimhood, but Muslim-majority countries in the Middle East uniquely exhibited comparably strong expressions of glorified sentiments of heroism. In contrast, pro-Israel hashtag activism was driven by villain-centric messages focusing on the topic of denunciation mixed with the rhetoric of victimhood. Contrary to popular characterizations of the discursive conflict as a parallel clash of victimhood or vengeful emotions, the results reveal stark asymmetries in how wartime militancy was rhetorically framed. I propose a power-centric interpretation of the observed rhetorical asymmetries, which I argue stem from diverging practical needs in a heavily unequal bilateral conflict. I also reflect on regional differences within the global Palestine solidarity movement amidst debates about the cultural praxis of subalternity that have intensified since the outbreak of the Gaza War.

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Science Advances

When trust building looks like surveillance: Public polarization toward Chicago policing

2026-07-17 · DOI 10.1126/SCIADV.AEB7575

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Abstract. Public debate about policing highlights competing views about whether community initiatives build trust or reinforce surveillance. We argue that perceptions depend on prior state orientations, which can polarize opinions across groups. Using a preregistered survey experiment in Chicago, we examined responses to the police department's "Sentiment Dashboard," a project designed to enhance legitimacy by collecting community feedback. Results show that the dashboard increased trust in police among non-Black respondents but reduced trust among Black respondents, especially when framed as monitoring. Among non-Black respondents, dashboard exposure also reduced support for the Black Lives Matter movement and racial equity policies unrelated to policing, such as diverse hiring and climate justice. Mechanism testing with qualitative evidence suggests that treated non-Black participants were more likely to regard police reforms as substitutes for racial equity policies,

rendering further measures unnecessary. These findings reveal how state legitimacy initiatives may instead widen racial divides and weaken coalitions for structural reform.

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Proceedings of the National Academy of Sciences

Whistleblowers can contain the unethical externalities of human-AI delegation

2026-07-16 · DOI 10.1073/PNAS.2536668123

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Abstract. Prior work using controlled principal-agent experiments suggests two risks from delegating tasks to AI systems: Human principals are more likely to request profit-maximizing misconduct from AI agents than from human agents, and AI agents are more likely to comply. Here we test whether third-party observers can contain the resulting harm. In an incentivized die-reporting paradigm, principals instructed either a human or an AI agent how strongly to prioritize profit over accuracy, creating potential financial harm to a charity. We first confirm, with human principals (N = 600) and three large language models as AI agents, that delegation to AI produces larger negative externalities than delegation to humans. We then study observers who could pay a personal cost to flag a principal's instruction, canceling the principal's gain in favor of the charity, as a laboratory analogue of whistleblowing. In this observer study (N = 300), the probability of flagging increased with how unethical the principal's request was, but did not depend on whether the request was directed to a human or an AI agent. Because principals made more unethical requests under AI delegation, flagging was more frequent under AI delegation. When combined with agent behavior, this increase in flagging fully neutralized the negative externalities of AI delegation in our experimental setting. These findings support institutional protections for whistleblowers as one potential organizational safeguard against the harms of human-AI delegation.

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Many AI analysts, one dataset: Navigating the agentic data science multiverse

2026-07-14 · DOI 10.1073/PNAS.2606495123

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Abstract. Empirical conclusions depend not only on data but also on analytic decisions. Many-analyst studies have quantified this dependence: independent teams testing the same hypothesis on the same dataset regularly reach conflicting conclusions. But such studies require costly human coordination. We show that fully autonomous AI analysts built on large language models (LLMs) can, cheaply and at scale, produce the analytic dispersion observed in human many-analyst studies. In our framework, each AI analyst independently executes a complete analysis pipeline on a fixed dataset and hypothesis; a separate AI auditor screens every run for methodological validity. Across three datasets, AI analyst-produced analyses exhibit substantial dispersion in effect sizes, P -values, and conclusions. This dispersion can be traced to identifiable analytic choices in preprocessing, model specification, and inference that vary systematically across LLM and persona conditions. Critically, the outcomes are steerable: reassigning the analyst persona or LLM shifts the distribution of results even among methodologically sound runs. These results highlight a central challenge for AI-automated empirical science: when defensible analyses are cheap to generate, evidence becomes abundant and vulnerable to selective reporting. The same capability also helps address it: treating analyst results as distributions makes analytic uncertainty visible, and deploying AI analysts against a published specification can reveal how much disagreement stems from underspecified design choices. Taken together, our results motivate a transparency norm: AI-generated analyses should be accompanied by multiverse-style reporting and full disclosure of the prompts used, on par with code and data.

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Combined signals of scientific and social consensus best mobilize action on societal challenges even if government and industry oppose

2026-07-13 · DOI 10.1073/PNAS.2533408123

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Abstract. When facing real-world societal challenges, people make decisions about how to act based on multiple, often conflicting messages from various influential stakeholders such as governments, scientists, fellow citizens, and private companies. How does the public decide to act amid potentially converging or diverging signals from multiple entities in society? Which entities remain influential even if other groups appear apathetic or stand in opposition? This question becomes especially important in contexts like the United States where the government, industry, scientists, and public sentiment may diverge. In two conjoint experiments spanning over 40,000 evaluations, one behavioral experiment and a digital field study (N s = 1,515, 2,005, 1,604, 50,668 respectively; all preregistered), we show the relative and combined influence of government bodies, scientific experts, industry stakeholders, and social consensus in the general public in driving responses to 10 issues in environmental, health, and technology domains. Across domains, support from scientists and social consensus was most influential in shaping the American public's responses to societal problems including policy support, product preferences, donations to a cause, and social media engagement. And when facing opposition from government and industry, the combined, but not isolated, support of scientists and social consensus reliably mobilized action across domains. This work suggests that even amid rising systemic opposition to solving contemporary societal challenges, scientists and the public can come together to mobilize action.

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American Political Science Review

Rebel Victory and Authoritarian (Dis)Order

2026-07-15 · DOI 10.1017/S0003055426101737

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Abstract. Many authoritarian regimes originating in violent rebellion are exceptionally durable. Yet others collapse quickly after their birth. We argue that the structure of the founding rebellion explains these divergent trajectories by shaping power-sharing dynamics in the new regime and the nature of conflict recurrence. Regimes founded by unified rebellions (single armed group) usually become highly durable autocracies because their members can credibly share power within the same armed organization. By contrast, regimes founded by fractured rebellions (multiple armed groups) often fail quickly because armed rivals must attempt to share power across their organizations, prompting defections and recurrent rebellions. Empirically, we support these arguments by analyzing originally compiled data on every regime founded by successful rebellion worldwide from 1900 to 2020. Neither alternative rebellion characteristics, like social revolution or leftist ideology, nor country-level factors can explain why rebel regimes follow divergent trajectories.

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Journal of Communication

Media capture, academia capture? Rising authoritarianism and public knowledge capture

2026-07-15 · DOI 10.1093/JOC/JQAG008

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Abstract. Abstract unavailable.

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Nature

Food systems transformation would reshape global agriculture

2026-07-15 · DOI 10.1038/S41586-026-10775-2

Matthew Gibson; Marina Sundiang; Daniel Mason-D'Croz; Thais Diniz Oliveira; Felicitas Beier; Lauren Benavidez; Astrid Bos; Maksym Chepeliev; Jonathan Doelman; Shahnila Dunston; Shinichiro Fujimori; Tomoko Hasegawa; Petr Havlik; Jordan Hristov; Jonas Jagermeyr; Marta Kozicka; Marijke Kuiper; Page Kyle; Thijs de Lange; Benjamin Leon Bodirsky; Hermann Lotze-Campen; Hermen Luchtenbelt; David Meng-Chuen Chen; Abhijeet Mishra; Christoph Muller; Gerald Nelson; Amanda Palazzo; Ignacio Perez Dominguez; Alexander Popp; Ronald Sands; Marco Springmann; Elke Stehfest; Timothy B. Sulser; Kiyoshi Takahashi; Gianmaria Tassinari; Ferike Thom; Philip Thornton; Kazuaki Tsuchiya; Willem-Jan van Zeist; Hans van Meijl; Dominique van der Mensbrugghe; Detlef Van Vuuren; Hannah H. E. van Zanten; Isabelle Weindl; Keith Wiebe; Xin Zhao; Mario Herrero

Abstract. Abstract Food systems are a major contributor to exceeding planetary boundaries 1-3 and poor quality diets are a key mortality risk globally 4 . Projected population and income growth could exacerbate these challenges 5 . In response, there are calls for transformation towards healthy and sustainable food systems 6-8 . However, the scale and distribution of the impacts of this transformation on agriculture are underexplored. Here we show that, by 2050, the transformation of food systems towards healthy diets (adoption of the EAT- Lancet reference diet), improved productivity and halving of food waste results in a fundamental restructuring of global agriculture, aspects of which break with historical trends. Scenario simulations using a multimodel ensemble of ten global economic models show a 6% median decrease in agricultural land (+1% to 26%) compared with 2020 levels. By 2050, agricultural production would be 17% lower than business-as-usual projections (2% to 32%) and, economically, the value of this production is US$1.6 trillion (26%) lower (+8% to 58%). Within this, the value of livestock production would be substantially lower than current 2050 projections (49% to 83%), while vegetable, fruit, nut and legume production value would increase by 23% (33% to +106%). Results are dependent on the assumed policies to achieve the transformation scenario. We highlight a more active role for food policy to consider the benefits of such a transformation (improved population health and reduced environmental pressures) and navigate the political economy of its impacts.

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Nature Communications

The economically optimal mix and timing of coastal adaptation in Europe to 2150

2026-07-15 · DOI 10.1038/S41467-026-74042-8

Vanessa Volz; Jochen Hinkel; Daniel Lincke; Lars E. Honsel; Robert J. Nicholls; Remi Thieblemont; Goneri Le Cozannet; Paul Sayers

Abstract. Abstract Coastal adaptation to sea level rise involves combining and sequencing adaptation options into adaptation pathways that include adaptation tipping points, which are moments requiring a switch between options, often due to economic limits. While local case studies show costs depend on the economically optimal timing of actions and adaptation tipping points, this remains unexplored on global and continental scales. We present economically optimal adaptation pathways for Europe's coastal floodplains through 2150. For 95% of the coastline requiring adaptation, the optimal timing for initial actions is immediate due to current economic under-adaptation, a condition in which the costs of additional adaptation are outweighed by the reduction in expected flood damages. By 2150, retreat is economically optimal for 22% of the coastline, incurring lower total costs than protection, which is optimal for 9%. Adaptation tipping points are primarily driven by sea level rise and necessitate a switch from accommodation to either protection or retreat, or from retreat to inland protection

combined with retreat in the foreland. Their timing depends on climate change, occurring 32 years earlier under higher climate change scenarios compared to lower ones.

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Late Additions

Nature

Intergenerational mobility fosters innovation in Europe

2026-07-08 · DOI 10.1038/S41586-026-10736-9

Sarah McNamara; Guido Neidhofer; Patrick Lehnert

Abstract. The circumstances into which individuals are born can place fundamental constraints on their future economic opportunities1-5, leading to a mismatch between talent, education and occupation. One major determinant of this inequality of opportunity is the absence of intergenerational educational mobility2. Here we extend existing knowledge on intergenerational mobility by presenting The European Atlas of Spatially Disaggregated Intergenerational Mobility (EUROPE-IGM-ATLAS), a panel database comprising indicators of intergenerational mobility for European subnational regions. In doing so, we make two contributions. First, we extend existing knowledge on the development of intergenerational mobility in European regions. The EUROPE-IGM-ATLAS reveals several spatiotemporal trends that characterize the changing geography of opportunity in Europe. For example, we show that observed increases in intergenerational mobility primarily stem from improvements in educational achievements among individuals from families at the lower end of the educational distribution, with fewer changes in rank across the educational spectrum. However, these increases are not uniformly distributed. Regions with a high degree of educational inequality also exhibit lower levels of intergenerational mobility, implying the co-existence of inequality both within and between generations. Second, we use this database to provide evidence on the relationship between intergenerational mobility and innovation. We provide large-scale time-series evidence that European regions with higher intergenerational mobility achieve higher innovation outcomes, one important driver of economic growth. Subsets of results further indicate that this relationship is nonlinear and that distinct mechanisms operate in major innovation hubs. The EUROPE-IGM-ATLAS reveals spatiotemporal trends that characterize the changing geography of opportunity in Europe and its relationship with regional innovation.

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